Pose Estimation Research

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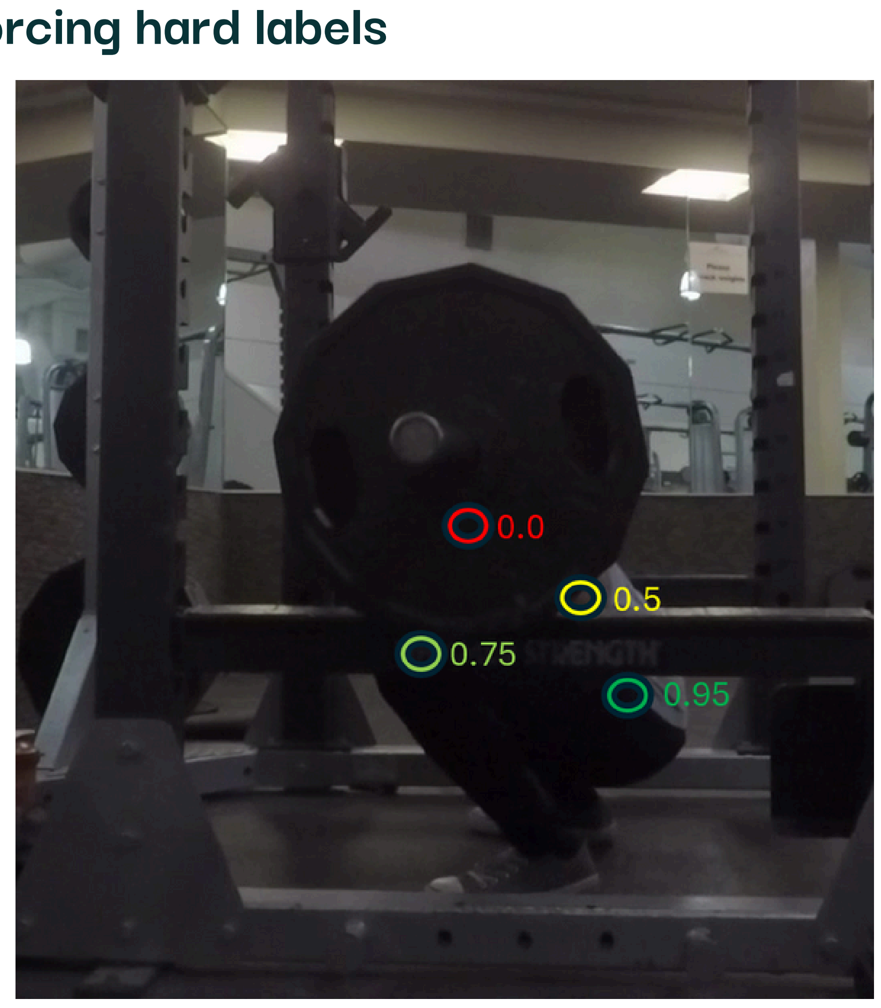
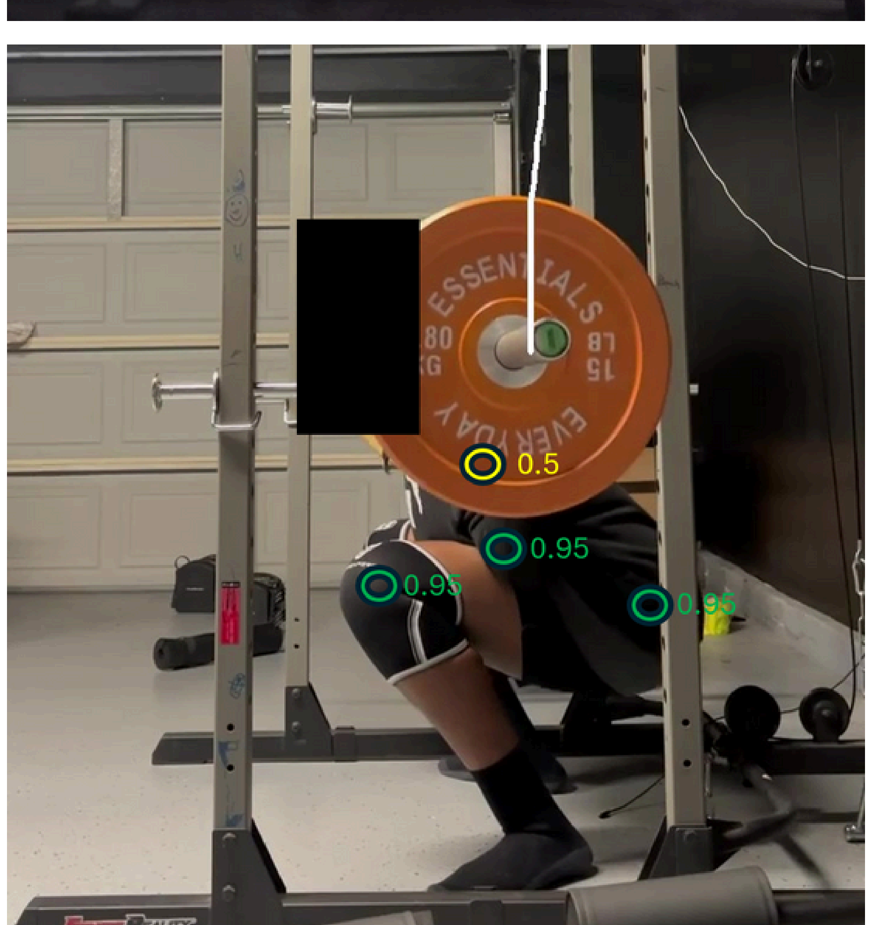
MOTIVATION

- Barbell squats frequently occlude shoulders and elbows with equipment, making rule-based form analysis difficult
- Pose estimation models such as OpenPose struggle to localize keypoints under even minor occlusion
- This project explores whether targeted fine-tuning can improve occluded keypoint prediction for lifting analysis

DATASET/ANNOTATION

- 3 datasets - train, validation, and test - were created
- Train dataset was created with annotated OpenPose predictions and custom confidences
- Validation and test sets created with CVAT
- Focus on left shoulder, left elbow, left hip, and left knee
- Confidence scores were assigned to encode uncertainty under occlusion rather than forcing hard labels





Representative frames demonstrating annotation and confidence criteria

RESEARCH QUESTION

Can a pose estimation model fine-tuned on a fixed side-view squat dataset learn to infer occluded shoulder and elbow keypoints from single frames?

MODEL/TRAINING SETUP

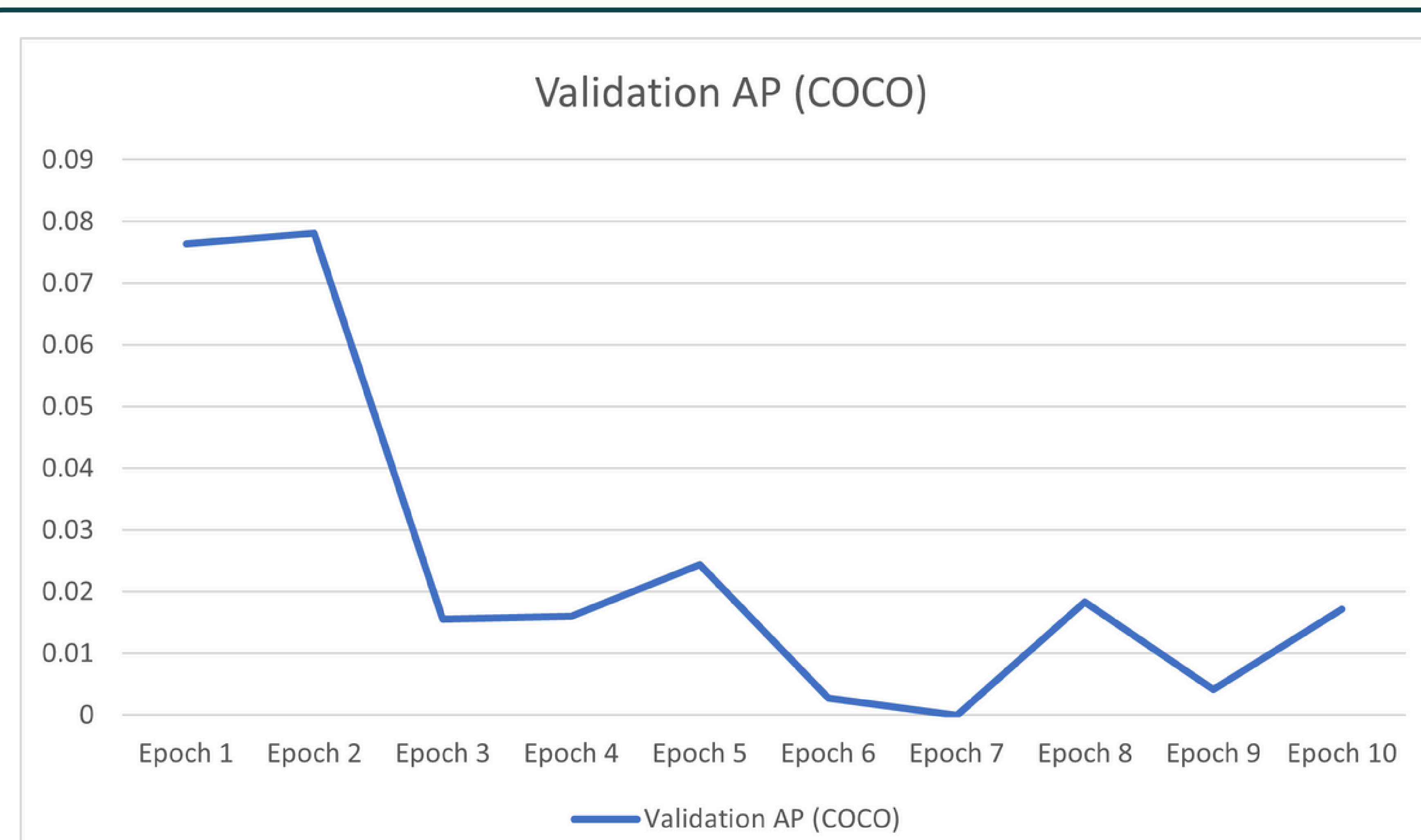
- Base model: HRNet-W18
- Standard COCO-17 keypoints changed to custom 4-keypoint setup
- 10 runs

RESULTS

Average Precision (AP) plateaued within the first two training epochs and did not improve with additional training, regardless of augmentation. AP and training loss were low from the start, indicating early convergence without meaningful learning. These results suggest dataset limitations rather than optimization issues.

QUALITATIVE RESULTS

- Limited structure in the predictions
- Predicted keypoints often placed on or near the subject's body, but positions varied inconsistently
- No meaningful anatomical pattern
- Same behavior for both occluded and non-occluded keypoints



LIMITATIONS

- Size and scope of the dataset
- Restricting the viewpoint narrowed the problem space, but limiting the task to only four keypoints reduced the model's ability to generalize across different human anatomies, lighting, and angles
- The annotations were manually generated and noisy despite consistent confidence rules

FUTURE WORK

- Develop a synthetic human pose dataset in Blender using a Mixamo rig with perfect ground-truth keypoints
- Systematically vary occlusion and pose to create evaluation conditions
- Evaluate HRNet-W32
- Quantify which factors most affect model performance